

power supplies, data centres, telecommunications equipment, and battery backup systems.

Link. <https://www.electronicsforu.com/electronics-projects/sic-based-polymorphic-dc-dc-converter-reference-design>

OEM Brand. Microchip

Bi-directional, dual active bridge

Texas Instruments' (TI) reference design supports battery charging and discharging through single-phase-shift (SPS) and extended-phase-shift (EPS) control while enabling soft



switching to reduce switching losses without auxiliary components. Its modular structure allows multiple converters to be combined for higher power levels. The converter delivers up to 10kW, with peak efficiency of 98.7% and 98% efficiency at full load. It operates with a primary voltage of 700V-800V DC and a secondary voltage range of 350V-500V DC under SPS or 250V-500V DC under EPS. The 100kHz switching frequency allows the use of a compact planar transformer. The design also includes isolated voltage and current sensing, a smart gate driver with integrated protection for silicon carbide MOSFETs, and a digital controller.

Applications. The reference design is intended for single-phase DAB converters used in DC fast chargers, energy storage systems, power conversion systems, and hybrid and electric vehicle powertrains.

Link. <https://www.electronicsforu.com/electronics-projects/>

bi-directional-dual-active-bridge-reference-design

OEM Brand. Texas Instruments (TI)

DC-DC converter with DAB topology

The reference design is a 5kW isolated bidirectional DC-DC converter for EV charging infrastructure and solar power systems. The design uses a dual active bridge (DAB) topology with full bridges



on both the high- and low-voltage sides, enabling bidirectional power transfer by controlling the phase difference between the two bridge circuits. Compared with half-bridge topologies, the full-bridge DAB configuration supports higher power levels and enables soft switching through phase-shift power transfer, reducing switching losses and improving conversion efficiency. By using SiC MOSFETs instead of insulated gate bipolar transistors (IGBTs), the design achieves higher switching frequencies, lower switching losses, and higher conversion efficiency than IGBT-based designs.

Applications. The reference design is suitable for EV charging stations and solar power inverters.

Link. <https://www.electronicsforu.com/electronics-projects/dc-dc-converter-reference-design-with-dab-topology>

OEM Brand. Toshiba

High-efficiency bidirectional DC-DC converter

Renesas' reference design for a bidirectional 12V/48V DC-DC converter transfers more than 3kW of power with over 95% efficiency,

enabling power to flow between 12V and 48V battery networks. It steps down power from the 48V battery to supply 12V electronics such as lighting, infotainment, ECUs, sensors, and safety systems, while also boosting the voltage from the 12V side to charge the 48V battery when required. This supports regenerative braking, battery management, and load balancing. Based on the LV148 48V automotive power-supply standard, the reference design uses a bidirectional analogue controller instead of a DSP-based control system, reducing software complexity, PCB size, bill-of-materials costs, and development time.

It also features a multi-phase synchronous DC-DC topology that improves power density, reduces thermal stress, and supports faster transient response, while integrating power devices, gate drivers, micro-controllers, voltage regulators, and protection circuitry.

Applications. The design supports dual-voltage power systems in electric vehicles, hybrid electric vehicles, mild hybrid electric vehicles, and electric two-wheelers.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-high-efficiency-bidirectional-dc-dc-converter>

OEM Brand. Renesas

These reference designs address different EV power conversion requirements using technologies such as dual active bridge topologies, bidirectional power conversion, and silicon carbide devices. Together, they demonstrate approaches for improving efficiency and power density while reducing development effort across EV charging, battery management, energy storage, and other power conversion applications. **EFY**

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